

Reviewer Pack — Minimal Rank of Quadratic Residues over Elliptic Curves vs B...

Entry 49483543ef97 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quadratic Residues over Elliptic Curves vs BP_ReadTwice Circuit Size

Entry ID: 49483543ef97 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 06:39:01 UTC

1. Conjecture

Field A (mathematical branch): Elliptic Curve Cryptography (Quadratic Residue Theory) **Field B** (complexity object): Complexity Theory: BP_ReadTwice Circuit Complexity

Statement:

{‘sentence_1’: ‘For a given input bit string x , the minimal rank of the elliptic curve over a finite field F_q that has x as a point with respect to quadratic residues is $O(\log q)$, where q is the size of F_q .’, ‘sentence_2’: “However, for a BP_ReadTwice circuit C computing an inner product mod 2, the minimal rank of the associated elliptic curve over F_q with respect to quadratic residues is $\Omega(n \log n)$, where n is the size of C ’s input.”, ‘sentence_3’: ‘This implies that distinguishing between BP_ReadTwice circuits for inner product and XOR requires a circuit complexity of $\Omega(n)$.’}

Rationale (proposer’s reasoning):

{‘sentence_1’: ‘Quadratic residue theory over elliptic curves provides a rich algebraic structure that could potentially be exploited to encode computational problems into geometric objects.’, ‘sentence_2’: ‘The BP_ReadTwice complexity is known to be a powerful model for capturing certain aspects of circuit complexity. The minimal rank of the associated elliptic curve may reveal hidden structures in these circuits.’, ‘sentence_3’: ‘The conjecture suggests that there exists a direct connection between the algebraic structure of elliptic curves and the complexity of BP_ReadTwice circuits, which could lead to new insights into computational separations.’}

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 823435928756f90d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the mean minimal rank difference between BP_ReadTwice circuits for inner product and XOR over all 30 seeds is less than or equal to $0.5 \log n$, where n is the input size of the circuits.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.90	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'quadratic residues' AND 'elliptic curve cryptography' AND 'BP_ReadTwice circuit' - 'minimal rank elliptic curve' AND 'quadratic residue theory' AND 'complexity BP_ReadTwice' - 'inner product mod 2' AND 'XOR' AND 'circuit complexity $O(\log q)$ ' AND ' $\Omega(n \log n)$ '

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def is_quadratic_residue(a, p):
        if a == 0:
            return True
        for i in range(1, p):
            if (i * i) % p == a:
                return True
        return False

    def find_elliptic_curve(x, q):
        while True:
            a = random.randint(1, q - 1)
            b = random.randint(0, q - 1)
            discriminant = (4 * a * a + 27 * b * b) % q
            if is_quadratic_residue(discriminant, q):
                return a, b

    def minimal_rank(a, b, q):
        rank = 0
```

```

    for i in range(q):
        if (i * i) % q == (a * i + b) % q:
            rank += 1
    return rank

n_values = [5, 10, 15, 20, 30, 40]
total_rank_diff = 0
instances_tested = 0

for n in n_values:
    for _ in range(5): # Ensure at least 5 instances per size
        x = ''.join(random.choice('01') for _ in range(n))
        q = random.randint(2**n, 2**(n+1)) - 1
        a, b = find_elliptic_curve(x, q)
        rank_inner_product = minimal_rank(a, b, q)

        # Construct BP_ReadTwice circuit for XOR (simplified example)
        rank_xor = n

        total_rank_diff += abs(rank_inner_product - rank_xor)
        instances_tested += 1

mean_rank_diff = total_rank_diff / instances_tested
conjecture_holds = mean_rank_diff <= 0.5 * math.log(n_values[-1])

return {
    "metric_name": "Mean Rank Difference",
    "metric_value": mean_rank_diff,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": "" if conjecture_holds else f"Rank difference {mean_rank_diff} > 0.5 * 1
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(30)] # Default to first 30

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"Rank difference exceeds threshold\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot determine if the conjecture is supported or falsified. | next: Re-run the test with increased time limits to ensure it completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17258
2	preregistration	ollama_remote	glm4:latest	0	0	5263
3	novelty	ollama_remote	glm4:latest	0	0	5012
4	novelty	ollama_remote	glm4:latest	0	0	5489
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11123
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8969
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7870
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9140
9	judge	ollama_remote	glm4:latest	0	0	11082

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 81204 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/49483543ef97.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/49483543ef97.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/49483543ef97.tar.gz` (if generated)